

Migratable urban street scene sensing method based on vision language pre-trained model

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ABSTRACT

We propose a geographically reproducible approach to urban scene sensing based on large-scale pre-trained models. With the rise of GeoAI research, many high-quality urban observation datasets and deep learning models have emerged. However, geospatial heterogeneity makes these resources challenging to share and migrate to new application scenarios. This paper introduces vision language and semantic pre-trained model for street view image analysis as an example. This bridges the boundaries of data formats under location coupling, allowing for the acquisition of text-image urban scene objective descriptions in the physical space from the human perspective, including entities, entity attributes, and the relationships between entities. Besides, we proposed the SFT-BERT model to extract text feature sets of 10 urban land use categories from 8,923 scenes in Wuhan. The results show that our method outperforms seven baseline models, including computer vision, and improves 15% compared to traditional deep learning methods, demonstrating the potential of a pre-train & fine-tune paradigm for GIS spatial analysis. Our model could also be reused in other cities, and more accurate image descriptions and scene judgments could be obtained by inputting street view images from different angles. The code is shared at: github.com/yemanzhongting/CityCaption.

1. Introduction

GeoAI methods represented by deep learning have made considerable progress in the field of GIS (Janowicz et al., 2020), especially in streetscape image analysis (Zhang et al., 2021a) and geotagged text analysis (Chen et al., 2021). However, GeoAI research also faces the urgent problem of weak Spatio-temporal replicability (Goodchild and Li, 2021). Unlike physical, biological, or computer disciplines, the presence of spatial dependence and spatial heterogeneity in urban studies leads to weak geographic replicability (Kedron et al., 2021). When changing the study area, although one may use the same data sources and the same analysis methods, it is not easy to guarantee that the same conclusions could be obtained. The occurrence and interpretation of spatial phenomena may also change due to the differences in architectural style, socio-economic, natural environment, and custom-culture in different cities (Xue et al., 2022; Kedron and Holler, 2022). To cope with such challenges, many scholars have worked on building open-source geographic databases (Nüst and Pebesma, 2021) and open-source geographic models (Wang et al., 2016).

Moreover, existing research has a small range of applicability, the geospatial knowledge mined with large amounts of computational power is difficult to share, and very little research has been on transfer learning with only a small amount of manually labeled data (Huang, 2022). Cities are the primary space for human activities, and with the increase in the number and variety of sensors, massive amounts of geo-tagged multimodal data are generated all the time (Zhang et al., 2020b). These data enrich our means of urban sensing and could be verified against each other (Wang, 2021). At the same time, the vast amount of multimodal data also poses many challenges.

The first is the “data bottleneck”. It is well known that deep learning is a data-driven approach, and models often include a large number of parameters. The cost of accurate and reliable large-scale data annotation is high, while few-shot learning often suffers from underfitting problems.

The second is the “computing power bottleneck” that occurs as the complexity of the model increases. The deep learning model layers are gradually increasing, and the model’s depth is gradually rising (Wang et al., 2022a). For example, GPT-3 includes 175 billion

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parameters (Dale, 2021), switch transformers include 1.6 trillion parameters (Fedus et al., 2021), the base version of Bidirectional Encoder Representations from Transformers (Bert) includes 110M parameters, the large version includes 340M parameter (Devlin et al., 2019a). Clearly, such large models require a dramatic increase in computing power, and ordinary users can hardly afford such high training costs.

Thirdly, there is the “model bottleneck”. We know that there are quite a few street view “downstream tasks” in the GeoAI field, such as city streetscape sentiment sensing (Yao et al., 2019), urban form (Guyot et al., 2021; Inoue et al., 2022), house price prediction (Law et al., 2019), human cognitive bias (Zhang et al., 2021c), land use classification (Cao et al., 2018; Fang et al., 2021), environment and health modeling (Helbich et al., 2019), etc. Although there is a considerable amount of open source code for deep learning models, there is still a lack of a relatively universal model that makes it difficult to generalize to other GIS tasks (Liu and Biljecki, 2022). Facing new geographic problems requires much resources to re-acquire data, annotate it, build models, and train them. In the meantime, the multi-source, highly dispersed, and heterogeneous nature of urban data makes them difficult to process, fuse and analyze. There is an urgent need for powerful multimodal models to bridge the boundaries between different modal data.

To solve the above three bottlenecks, we introduced a large-scale vision language pre-trained (image caption) model and semantic representation (Bert) model. Our approach has two main advantages. First, it could improve the replicability of GIS models and is actually a highly semantic compression of urban landscapes. It could extend the environment sensing of streetscape into the image-text sensing of the scene. Second, a certain degree of resource sharing is achieved. Using a pre-trained vision language model could reduce computational costs and carbon footprint, allowing state-of-the-art (SOTA) models without training them from scratch.

As far as we know, this is the first application of a large-scale vision language pre-trained model in the spatial analysis of street images. It bridges the data gap between geo-tagged text and street view images. In other words, it could objectively translate the physical urban space into human language. This could avoid the subjective descriptions in geographic text data and the inter-city differences brought by the architectural styles (Sun et al., 2022) in geographic images.

What is more, we find that fine-tuning (through constructing appropriate loss functions) the relevant and suitable pre-trained model for a specific geographical task (the downstream task in this paper is urban scene classification based on street view images) could achieve SOTA results with less computational power cost and less labeled data.

The remainder of the paper is organized as follows: Section 2 compares existing research on pre-trained models and street view image analysis, leading to our work. The third section introduces the vision language pre-trained (image caption) model, the semantic representation model Bert, and our fine-tuned model SFT-BERT. Section 4 is our experimental part, where we reason about the urban street view and obtain human language representations of the scene, on which we perform unsupervised and supervised analyses, respectively. The SFT-BERT model could easily achieve SOTA results. The article ends with a conclusion and outlook in Section 5.

2. Literature review

Pre-trained models are models trained on publicly available large-scale datasets created by someone else, but could be used to solve similar problems. Instead of building models from scratch and focusing on the cold starts, users could use relevant trained models as a starting point for knowledge-based relearning (Han et al., 2021). With the powerful generalization capabilities, large-scale pre-trained models have made significant progress in natural language processing (NLP) and multi-modality areas (Qiu et al., 2020).

A large number of open-source pre-trained techniques and models have emerged, such as ViBERT (Lu et al., 2019) and UNIMO (Li et al., 2021) for handling dozens of tasks such as visual question answer, Word2vec (Mikolov et al., 2013) and Glove (Pennington et al., 2014) for learning shallow semantic features of word co-occurrence probabilities, Bert (Devlin et al., 2019b), ELMo (Peters et al., 2018) and GPT (Brown et al., 2020) with contextual semantic representation of word embeddings, which have greatly contributed to the development of related fields.

The more common application of pre-trained models is in image processing, such as remote sensing (Qin and Liu, 2022). Those vision pre-trained model VGG (Simonyan and Zisserman, 2015), Inceptionv3 (Szegedy et al., 2015), ResNet50 (Fang et al., 2021; He et al., 2016), EfficientNet (Tan and Le, 2019), etc. are trained on open-source datasets to obtain classification capabilities with up to thousands of labels. These models naturally have high accuracy in scenarios such as land use classification and elemental object recognition in remote sensing imagery (Law et al., 2020). Unlike remote sensing image, which has a regular format, GIS data is more complex in format, mostly in the form of text and graphics (Namouchi and Farah, 2021). Most current researches of pre-trained models in the GIS domain focus on direct calls to fixed representations of encoders, with applications in land use type classification (Word2vec) (Yao et al., 2017; Zhang et al., 2022b), urban function recognition (Glove) (Zhang et al., 2021d), social media text encoding (Bert) (Qian et al., 2020; Yao and Wang, 2020), flood address extraction (Bert) (Zhang et al., 2021b), short text structure recognition (Bert) (Liang et al., 2021; Hu et al., 2022; Zhang et al., 2020a) and other domains. Very few studies have noted multimodal pre-trained models in GIS and its powerful Few-Shot learning capability (Zhang et al., 2022a). In fact, the pre-trained and fine-tuned architecture could be applied to most GIS tasks, such as tweet sentiment analysis, streetscape sentiment classification, address text recognition, and intelligent knowledge Q&A. Compared to pre-train, fine-tuning requires only a small amount of computational resources and fewer data to achieve satisfactory results. Besides, the associated suitable pre-trained model is reusable. This paper uses the urban street view as an example, and the analysis is based on this paradigm.

Street view data has proven to have good potential for application in the field of urban sensing (Kang et al., 2018; Weichenthal et al., 2019; Biljecki and Ito, 2021), containing a large amount of fine-grained information related to human (Zhang et al., 2018a). Its Applications include urban pollution monitoring (Apte et al., 2017), carbon emission estimation (Liu et al., 2021), human behavior pattern analysis (Yao et al., 2021), street information extraction (Wojna et al., 2017), identification of demographic composition (Gebu et al., 2017), scene sensing (Zhang et al., 2018b), and so on.

However, most of the existing work mentioned above only focuses on the correlation between urban street scenes and environmental variables in the study area. Both the semantic segmentation (Zhang et al., 2021a) and the objective detection (Wang et al., 2022b) of the street view could not describe the urban environment from the perspective of human language. This urgently requires new GIS methods that capture not only the spatial relationships between urban entities but also the key attributes of the entities and organize them logically in a holistic scene. To test whether the pre-trained model can be used effectively in GIS with less annotated data, we propose the SFT-BERT method based on the heterogeneity of text descriptions of different scenes. Compared with direct image classification of street scenes using convolutional neural networks (Srivastava et al., 2018, 2020), we extracted the human natural description features. It could obtain a “sense of scene” (Larkin et al., 2021; Zhang et al., 2020c), closer to subjective human sensing.

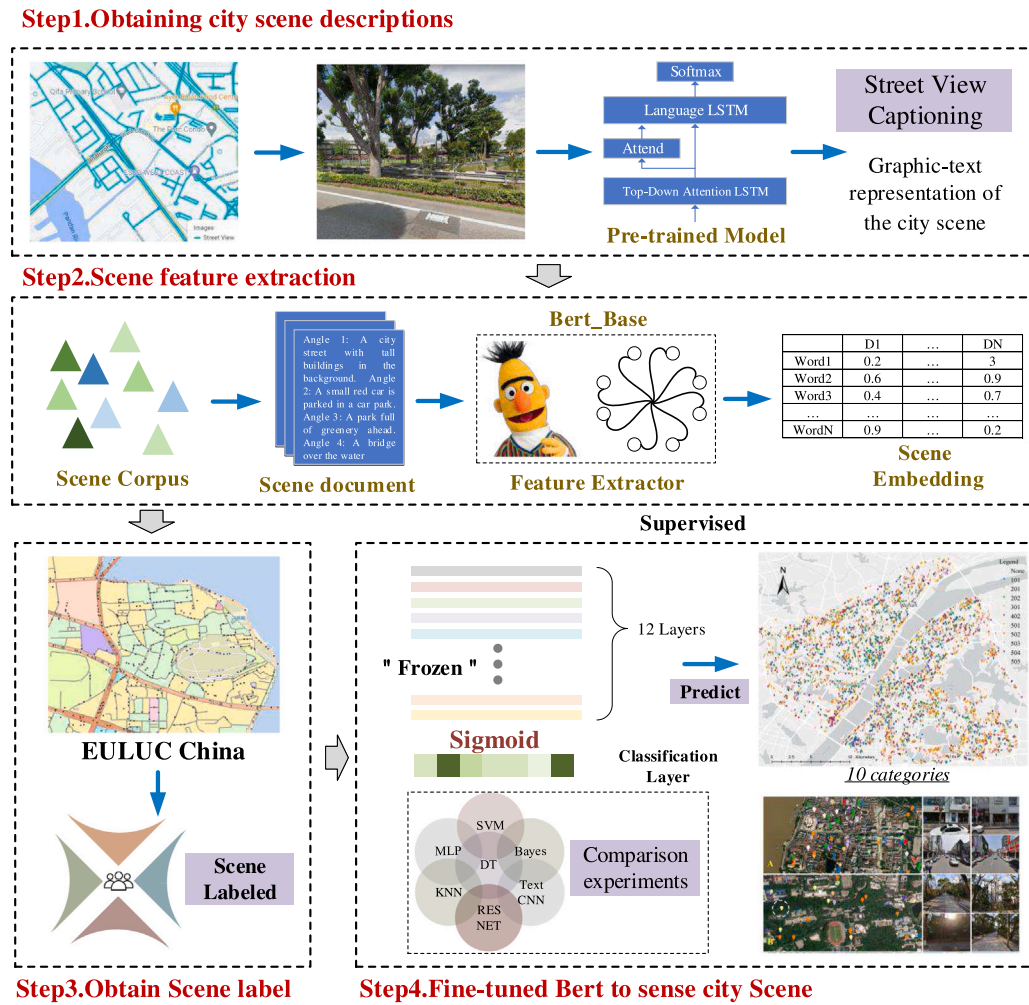


Fig. 1. General structure of this article (Step 1, input the street image and output the text description. Step 2, input the text description “corpus” to output the sentence features. Step 3, input EULUC China data, “inflate” the sample points, and get the labels. Step 4, input the sentence features and labels, fine-tune the model, and output the predicted results).

3. Research methods

As shown in Fig. 1, we divided this study into four main steps. In the first step, we reason about the street view scene description based on the pre-trained vision language model. It could perform highly semantic compression of street view images to obtain the geotagged text (Bahrehdar et al., 2020). In the second step, a sampled location will contain a few images from different angles. We aggregate them, and use Bert to get the initialized overall scene embedding. In the third step, the scene is labeled based on the Essential Urban Land Use Categories (EULUC) dataset. In the final step, we “freeze” the deep parameters of the Bert model, add a classification layer, and train it with the shallow parameters. We use the fine-tuned “SFT-Bert” model to predict urban scenes and evaluate them in comparison with the other six baselines.

3.1. Vision language model for image captioning

Image captioning is the process of generating textual descriptions for a image that could build a bridge between image modal and text modal. We need to deal with both text and image, which could be considered as a task of pixel sequences of images to word sequences of text. A common image caption model architecture is shown in Fig. 2, divided into two parts: encoder (green) and decoder (red).

For the decoder part, the most common one is the Long short-term memory (LSTM) recurrent neural network; for the encoder part

(green), there are three main methods: Resnet101 feature, Bottom-Up feature (Anderson et al., 2018; Rennie et al., 2017; Lu et al., 2020), and the vilbert feature (Lu et al., 2019). The main difference between the three encoding methods is that Resnet101 is a 101-layer-deep convolutional neural network trained on more than one million images that could identify 1,000 object classes. Bottom-Up method uses Faster R-CNN to extract image features, dividing the image into k regions. Vilbert feature extracting method is a joint representation model for images and natural language.

Annotation of multimodal data is extremely expensive, and the most prominent open source image caption datasets are as follows:

1. Microsoft COCO dataset (Chen et al., 2015) (It is the most commonly used training resource for multi-modality, including indoor and outdoor environments, with over 1.5 million captions describing over 330,000 images).
2. Flickr30k dataset (Young et al., 2014) (It contains 31,000 images collected from Flickr platform, and 5 reference sentences provided by human volunteer, which are mainly from social media).
3. Google Conceptual Captions dataset (Sharma et al., 2018) (3 million images, paired with natural-language captions, mainly from Internet HTML pages).

The Microsoft Common Objects in Context dataset is the most widely applicable and is the one we use. This dataset consists of indoor and outdoor scene, 80 object classes, and 5 manually annotated descriptions for each image. Microsoft company invites volunteers to provide textual descriptions of every images and to cross-validate them.

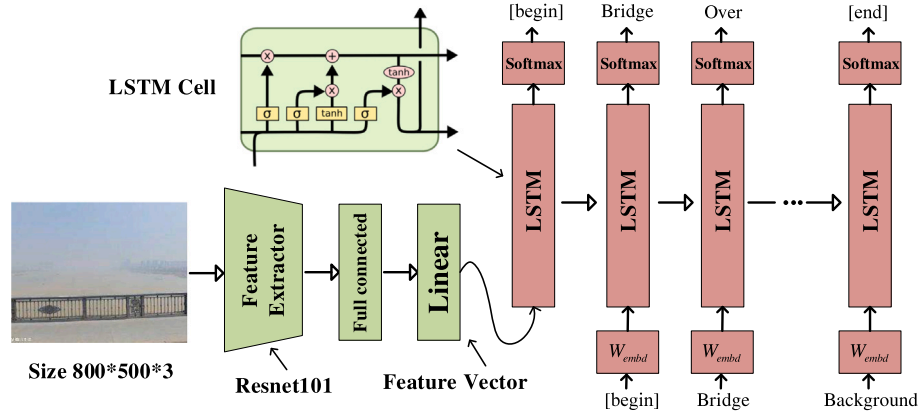


Fig. 2. Structure of the vision language pre-trained model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In the training process of the Image caption model, we input the images ($size = 800 \times 500 \times 3$) and the corresponding descriptions ($tokennumber = N$) simultaneously. We will add two placeholders [start] and [end] before and after the description to indicate the beginning and end, respectively. When the decoder module outputs [end] at the t step, it will indicate the end of the loop.

The model inputs the word embedding sequence $(x_1, x_2, \dots, x_N)$, computes the hidden sequence $(h_1, h_2, \dots, h_N)$, and obtains the output sequence $(y_1, y_2, \dots, y_N)$. The computational procedure shown in Fig. 2 is described by the following equation:

$$\begin{aligned} x_{-1} &= Encoder(image), x_t = Embedding_{x_t}, t \in \{0, \dots, N-1\}, \\ h_t &= LSTM(x_t, h_{t-1}), t \in \{0, \dots, N-1\}, \\ y_t &= p(next\ word | previous\ words) = softmax(W_p h_t^2 + b_p). \end{aligned} \quad (1)$$

where W_p are the parameters we need to train, b_p are the weights and biases to be learned, and the output y is the word i.e. the scene description of the street view image. After the model has reached convergence, we can save the parameters and provide a pre-trained model for downstream tasks. The main quality assessment index of the image caption model are CIDEr score (Vedantam et al., 2015) and SPICE score (Anderson et al., 2016). In this research, we calculate scene descriptions (geotagged-text) from street scenes (geotagged-image) using the pre-trained model (CIDEr=1.158 SPICE=0.2114) provided by Luo et al. (2018).

3.2. Structure of the bert semantic representation model

Bert is a Transformer neural network-based language model in the field of natural language processing (NLP), first proposed by Google in 2018; Bert and its structural variants refresh the optimal accuracy in 11 NLP tasks such as intelligent question and answer, semantic retrieval, sentiment classification, etc., and is one of the most advanced models in NLP. In this paper, we use the base version of Bert, which consists of 12 transformer layers (Vaswani et al., 2017), 12 attention heads, 768 dimensions, and 110M parameters. Pre-training is an unsupervised learning process on a large-scale corpus (such as wiki), where the model has three input layers. The first is the word embedding layer, which maps the text to the mathematical space. The second layer is the segment embeddings, wh to mark different sentences in the input model. And the third layer is the position embedding. Unlike traditional sequence models, the transformer used by Bert allows for parallel computation of the data, and the addition of the positioning layer effectively preserves the positional relationships between words.

In addition to words, [CLS] and [MASK] annotations are added during the pre-training of the model. The [CLS], as a symbol with no apparent semantic information at first, is used in the training process and is a more “fair” way to integrate the semantic information of the

total words in the text, making it an effective way to extract sentence features.

As shown in Fig. 3, Bert based his self-supervised training on large-scale text data (Wikipedia) on the Masked Language Model (MLM) and Next Sentence Prediction (NSP) tasks. The MLM task randomly masks 15% of the text input and uses the surrounding words to predict the masked words [MASK]. The NSP task is to input the sentence E_B and a randomly selected sentence E_{Random} from the corpus and let the model learn which one is the E_A following sentence (segment embedding, Fig. 3a). MLM teaches BERT to understand relationships between words. NSP teaches BERT to understand longer-term dependencies across sentences.

Pre-training a large model is very expensive and requires a lot of Graphics Processing Unit (GPU) computing resources (Wang et al., 2022c). On the contrary, we could “fine-tune” the pre-trained model based on the downstream tasks we need to implement for urban scene classification. “Fine-tuning” means using the Bert network as a feature extractor for the new dataset, keeping the weights of the initial layers of the model frozen and only retraining the shallow layers. This has the advantage of achieving SOTA results without requiring too much computational power and also performs good on few-shot dataset.

3.3. Fine-tuning the pre-trained models with scene sensing loss functions (SFT-BERT)

The pre-train & fine-tuning paradigm refers to learning general language features that are not directly relevant to the downstream task by pre-training, and learning features applicable to a specific task by fine-tuning. In general, in neural networks, the bottom and middle layers usually represent generic features, while the last few layers represent specific features (Er et al., 2016). The Bert-Base pre-trained model contains 12 transformer layers and 110 million parameters, when freezing the first 8 layers (i.e. tuning only the last 4) (Turc et al., 2019), the accuracy of the downstream task is not significantly reduced. In other words, we “freeze” most of the parameters of the Bert model and update the remaining parameters by adding new functional layers and auxiliary loss functions according to the specific task (Fig. 3b). And it could achieve state of the art (SOTA) results with much lower computing power than pre-training (Yu et al., 2020). The difference between fine-tuning and pre-train is that new functional layers are added after the last transformer encoder, and secondly, the text is no longer random “Masked”, as shown in Fig. 3.

The downstream task we need to solve is urban scene classification. We added sigmoid classification as a functional layer, after the Bert last transformer layer. It computes K class labels independently and K class losses are summed as the model total loss.

The reason for not using the common softmax cross-entropy loss function is that some areas in the city often have multiple functions.

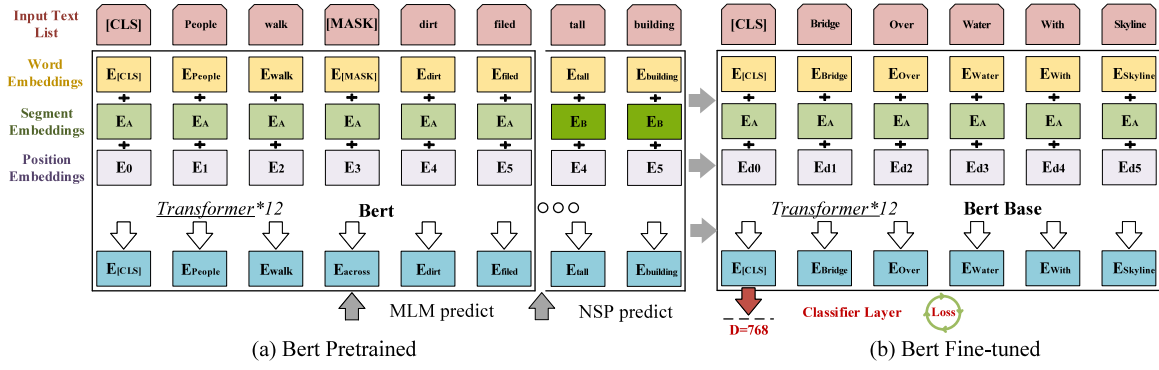


Fig. 3. The principle of Bert pre-training and fine-tuning (pre-training of a language model could take dozens of weeks to compute, whereas the Bert fine-tuning model only takes a few hundred seconds to run on a single GPU).

That is, these areas have the characteristics of multiple scenes at the same time and cannot be exactly classified into a certain category. If those locations with obscure features are forced to predict the appropriate labels, it will instead cause a decrease in accuracy.

The Sigmoid calculation equation is given by the following :

$$\text{sigmoid}(x_j) = \sigma(x_j) = \frac{e^{x_j}}{1 + e^{x_j}} \quad (2)$$

the original output of the model is x_j , $\sigma(x_j)$ is the probability that the output is of class j , and the value is between 0 and 1. For the model optimizer, we have chosen Bert's native Adam Weight Decay Optimizer, and the loss function equation is as follows:

$$\text{loss} = z * -\log(\text{sigmoid}(x)) + (1 - z) * -\log(1 - \text{sigmoid}(x)) \quad (3)$$

z is a list of data labels in the format $[0, 1, 0, \dots, 0]_K$, and x is the original output list of the model.

We named the scene sensing fine-tuned model as SFT-BERT. Referring to the Chinese urban functional product generated by article (Gong et al., 2020; Ma and Long, 2020), this paper divides the urban scenes into 10 classes ($K = 10$). If the model calculates $\sigma(x_i) > 0.2$, then marks class i as the label of the data. Since the data include multiple class labels, we use a quantitatively weighted model accuracy assessment method.

$$\text{Weighted-Precision} = \sum_{i=1}^K \beta_i \cdot \frac{TP_i}{TP_i + FP_i} \quad (4)$$

$$\text{Weighted-Recall} = \sum_{i=1}^K \beta_i \cdot \frac{TP_i}{TP_i + FN_i} \quad (5)$$

$$\text{Weighted-F}_1 = \sum_{i=1}^K \beta_i \cdot \frac{2 \times \text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (6)$$

Where β_i is the proportion of class i in the total data, True Positive (TP_i) denotes the number of positive samples of type i that were successfully predicted as positive, True Negative (TN_i) represents the number of negative examples of class i that were successfully predicted as negative, False Positive (FP_i) denotes the number of negative samples of type i that were incorrectly predicted as positive, False Negative (FN_i) denotes the number of positive samples of class i that were incorrectly predicted as negative.

4. Analysis of experimental results

Street view image is a geotagged city street-level photograph with the advantage of being publicly available and easily accessible (Liu et al., 2016). It is widely used in GIS for perception and quantitative analysis of cities (Kang et al., 2020). Our study area is within the 3rd Ring Road of Wuhan, Hubei Province, as shown in Fig. 4. The 3rd Ring Road of Wuhan is an urban area with the typical characteristics of an urban physical environment. A total of 75,628 street view images

(800*500 pixels) were obtained from 18,907 sampled locations in Tencent Map,¹ with 4 street view photos taken from different angles at each location.

Table 1 presents the data details. The perspective experience column consists of two types of values, a value of 2 is a machine sampled perspective with uncertainty, and a value of 4 is a manually calibrated perspective with better visibility of surrounding points of interest (POI). Angle column means the horizontal angle between the sampled view and due north, measured in degrees, while Picture Id is a combination of Area Id, orientation, and angle.

4.1. Get image-text representation of the city scene

We define the sensing of the surroundings at the location from different angles as an urban scene. Cities tend to produce similar street view images, but the different angles of the location image-text descriptions together build the scene characteristics. It could allow a more comprehensive reproduction of the actual scene.

We used the vision language model mentioned in Section 3.1 to perform scene inference on 75,628 street scene images of Wuhan city. The model generates text for each image, including the most relevant five sentences of street scene descriptions. A total of 378,140 text descriptions of the city were generated in about 72 h of reasoning on the Tesla P100 GPU. The best relevant descriptions from four different angles were stitched together to obtain a general textual description for representing the urban scene at the sampled location. We randomly selected eight locations in the study area. Then we listed the street view images and the best relevant image captions in Fig. 5. The only error occurs when the shadow of a pedestrian is incorrectly identified as a skateboard in Fig. 5G, but this does not affect the overall judgment of the scene.

This section automatically performs scene inference on street images based on the vision-language pre-trained model introduced in Section 3.1 and generates geo-tagged text descriptions. This advantage includes not only the elemental entities (which can also be achieved using common entity detection frameworks such as Mask CNN) but also the spatial location relationships and attributes between the entities. This could provide more intuitive knowledge than a geographical image. As the work in this section has extracted the key elements of scenes that are most interesting to humans, such feature engineering is also helpful for rapidly deploying scene recognition algorithms. The dataset could also be reused for related urban multimodal studies and has been uploaded and shared.

¹ <https://map.qq.com/>

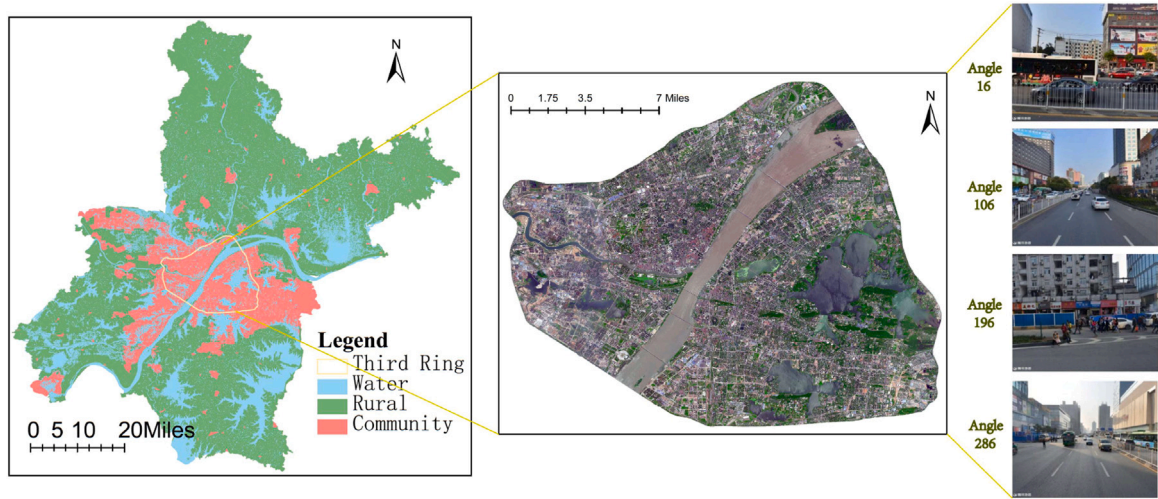


Fig. 4. Description of the study area and the collected data.

Table 1
Introduction to Street View Data.

Road Name	Latitude	Longitude	pov_exp	Angle	PictureId
S113	30.461	114.302	2	142	10141003150306134427600_back_142.jpg
S113	30.461	114.302	2	322	10141003150306134427600_front_322.jpg
S113	30.461	114.302	2	232	10141003150306134427600_left_232.jpg
S113	30.461	114.302	2	52	10141003150306134427600_right_52.jpg
...	...	...	...	...	...
HengAn	30.504	114.308	2	184	10141003150306135829100_back_184.jpg
HengAn	30.504	114.308	2	4	10141003150306135829100_front_4.jpg
HengAn	30.504	114.308	2	274	10141003150306135829100_left_274.jpg
HengAn	30.504	114.308	2	94	10141003150306135829100_right_94.jpg

Pov_exp: Perspective Experience.



Fig. 5. Scene descriptions of urban street views.

4.2. Exploring independent urban scenes with semantic information

To extract some deeper features of the urban scenes, the 18,907 scene description texts obtained by Section 4.1 were processed using the Bert sentence embedding method mentioned in Section 3.2 ([CLS] for 768-dimensional coding, Fig. 3b). We extract the semantic features of all scenes and generate a feature extraction matrix of size 18907×768 . This matrix was then clustered unsupervised using the HDBSCAN algorithm to discover how many separate scenes in the

city. The scene semantics was then reduced using Uniform Manifold Approximation and Projection (UMAP), which has the advantage of preserving as much of the global structure as possible, as shown in Fig. 6.

Fig. 6 shows that scenes can be better divided into 12 categories, which could be well distinguished from each other based on semantic features. In other words, the urban scenes have distinct category features. To uncover the entity-related relationships specifically contained in each class. We use the Latent Dirichlet Allocation (LDA) topic

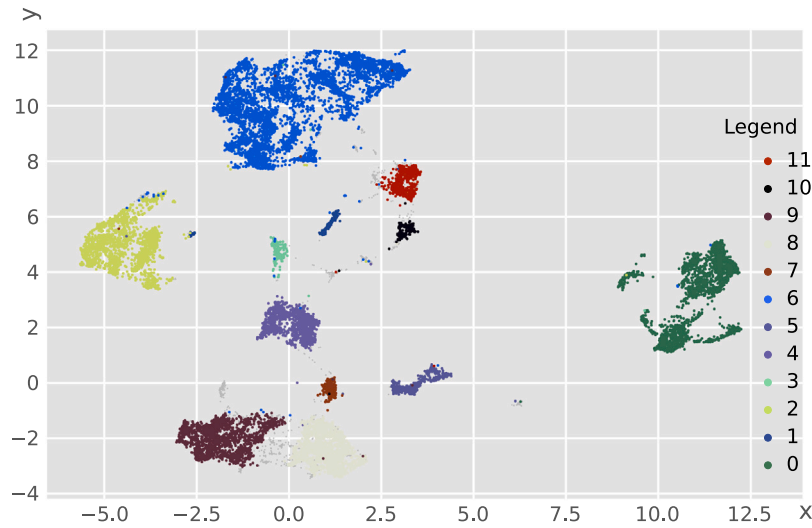


Fig. 6. Low-dimensional visualization of different categories of urban scenes after Bert sentence encoding.

Table 2

Frequency of co-occurrence of topic-words in urban scenes (we set the number of topics to the 12 categories calculated by HDBSCAN, and the ten words with the highest probability in each topic were showed).

TopicID	Topic0	Topic1	Topic2	Topic3	Topic4	Topic5	Topic6	Topic7	Topic8	Topic9	Topic10	Topic11
Word	“street”	“street”	“building”	“street”	“road”	“front”	“parked”	“road”	“lot”	“people”	“driving”	“street”
Prob	0.171	0.14	0.061	0.166	0.087	0.13	0.157	0.104	0.072	0.074	0.15	0.16
Word	“city”	“city”	“tree”	“city”	“parked”	“parked”	“side”	“trees”	“parking”	“walking”	“street”	“cars”
Prob	0.081	0.104	0.048	0.127	0.08	0.079	0.112	0.073	0.068	0.069	0.144	0.088
Word	“person”	“buildings”	“car”	“view”	“side”	“building”	“road”	“street”	“parked”	“group”	“cars”	“group”
Prob	0.049	0.087	0.045	0.088	0.075	0.074	0.09	0.053	0.062	0.066	0.087	0.086
Word	“view”	“traffic”	“next”	“car”	“car”	“store”	“building”	“dirt”	“cars”	“street”	“buildings”	“people”
Prob	0.046	0.058	0.043	0.048	0.066	0.068	0.079	0.043	0.059	0.066	0.063	0.074
Word	“riding”	“filled”	“street”	“driving”	“building”	“street”	“front”	“truck”	“building”	“bridge”	“next”	“city”
Prob	0.041	0.058	0.041	0.043	0.056	0.049	0.059	0.039	0.055	0.063	0.059	0.068
Word	“buildings”	“tall”	“middle”	“building”	“front”	“cars”	“car”	“next”	“couple”	“sidewalk”	“tall”	“walking”
Prob	0.035	0.055	0.041	0.029	0.047	0.043	0.058	0.036	0.05	0.053	0.051	0.049
Word	“walking”	“building”	“white”	“white”	“street”	“car”	“white”	“driving”	“wall”	“building”	“city”	“parked”
Prob	0.034	0.048	0.04	0.028	0.045	0.032	0.042	0.034	0.049	0.037	0.05	0.044
Word	“sidewalk”	“background”	“sidewalk”	“buildings”	“large”	“group”	“street”	“forest”	“group”	“view”	“car”	“sitting”
Prob	0.032	0.048	0.04	0.028	0.038	0.031	0.042	0.023	0.04	0.032	0.048	0.04
Word	“building”	“lots”	“sitting”	“across”	“long”	“buildings”	“cars”	“fence”	“street”	“water”	“group”	“couple”
Prob	0.03	0.028	0.037	0.027	0.035	0.029	0.042	0.023	0.035	0.031	0.035	0.024
Word	“next”	“empty”	“driving”	“skyline”	“cars”	“tall”	“truck”	“side”	“people”	“body”	“bus”	“side”
Prob	0.027	0.028	0.034	0.022	0.032	0.027	0.04	0.023	0.027	0.026	0.025	0.021

model to extract keywords under different scenes. LDA is a probability distribution model and does not care about the word’s location in the document. We consider all city scene descriptions as a Document and analyze their dependencies (Topic). The topic number is aligned with the number of clusters in HDBSCAN. We computed the topic-word co-occurrence frequency matrix from a corpus of 18,907 scenes containing 678 core words, and the results are shown in Table 2.

The results of topic mining seem to relate to a combination of scenes with elements such as roads, pedestrians, motor vehicles, etc. Yet we have difficulty in mining the physical meanings represented by each type of scene. In other words, the recognition results of these unsupervised approaches, both for the clustering of scene embeddings and the topic models, do not exactly match the standard urban functional classification criteria, which requires new supervised classification algorithms (Zhang et al., 2021b). In order to make our model directly callable in other cities, we built and trained our fine-tuned model in the next section, supplemented with standard city function classification labels.

4.3. Obtaining scene labels and predicting them using the SFT-BERT model

To make our model geologically meaningful, we used the EULUC-China dataset based on multiple data sources, with 12 different urban parcel attributes segmented by open street map data (Gong et al., 2020). The labels refer to the Chinese land-use status classification standard (GB/T 21010-2017).

Sorted by code in order, 101:residential category, 201: business office category, 202: commercial service category, 301: industrial category, 401: Road category (we do not care about this type), 402: transportation stations category, 403: airport facilities category, 501: administrative category, 502: educational category, 503: medical category, 504: sport and cultural category, 505: park and greenspace category.

As most street view images are captured by acquisition vehicles, the locations are often on urban roads. The width of urban roads is usually no more than 100 meters (Chen et al., 2019), so we set a buffer distance of 50 meters to “inflate” the street view locations and obtain labels for the urban scenes. In the study area, we did not care about the Road category, and the number of airport facilities category was only 1, so we only kept ten labels.

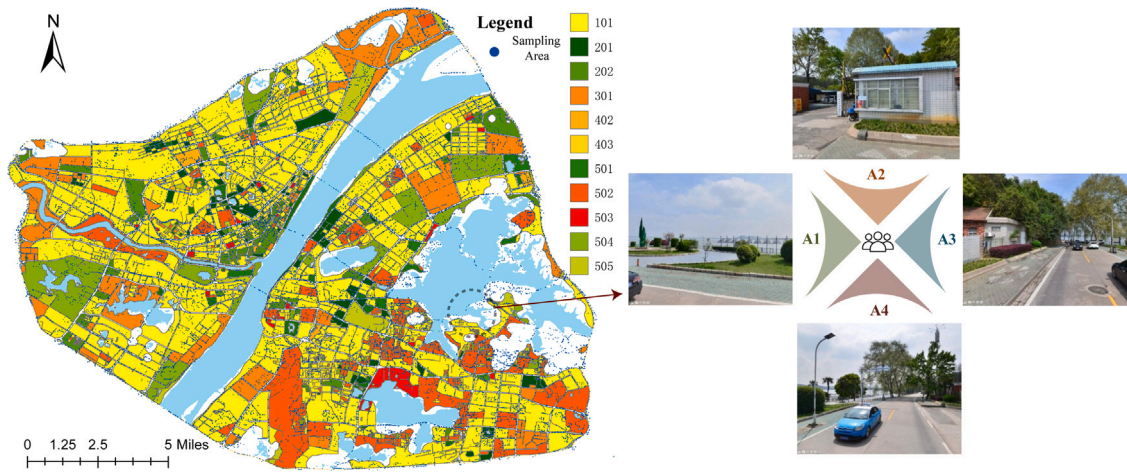


Fig. 7. EULUC functional classification schematic (descriptions of different angles are stitched together as a total scene description of the location, street view locations create buffers to obtain labels).

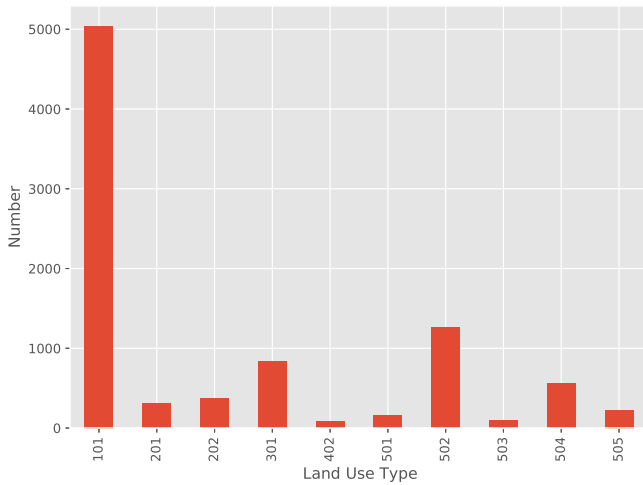


Fig. 8. Introduction to data distribution.

As shown in Fig. 7, stitching of the descriptions of the different views to represent the scene. A portion of the sampled locations did not have EULUC tags and was cleaned to obtain 8,923 meaningful scene locations. The distribution of the different label categories is shown in Fig. 8. This is clearly an acceptable amount of data compared to the hundreds of thousands of manual tags used in the pre-training process.

We use the SFT-BERT model introduced in Section 3.3, with the parameter settings referenced from the paper (Devlin et al., 2019a), BATCH_SIZE = 32, LEARNING_RATE = $2e-5$, MAX_SEQ_LENGTH = 128. The training set is 90%, the test set is 10%. The pre-training model is Bert_base_uncased_L-12_H-768_A-12 (Turc et al., 2019). The experiments were computed on a Tesla P100 GPU using the TensorFlow 1.15 deep learning framework. The trained model could calculate the probability of the data belonging to each urban scenario.

We compare SFT-Bert with the standard baseline algorithms TextCNN (Guo et al., 2019; Kim, 2014; Porzi et al., 2015), Multi layer Perceptron (MLP), K- Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree (DT), Naive Bayes classifier (Bayes), were compared in terms of precision, F_1 , recall, compute units, and model training times.

We trained SFT-BERT for three epochs and achieved SOTA results in 7 min and 21 s. Compared to the best-performing baseline algorithm, the model accuracy was improved by about 15%. The comparison

Table 3

Comparison results of SFT-BERT with the baseline algorithm.

Model	Weighted Precision	Weighted F_1	Weighted Recall	Cost Time (s)	Compute Unit
SFT-BERT	0.574	0.568	0.658	441.2	●
TextCNN	0.440	0.486	0.579	18.6	●
KNN	0.439	0.465	0.546	0.6	●
SVM	0.336	0.426	0.58	14.8	●
Decision Tree	0.400	0.402	0.404	1.5	●
MLP	0.421	0.443	0.471	15.0	●
Bayes	0.498	0.446	0.581	0.1	●
Resnet	0.319	0.408	0.565	963.3	●

Note: ● represents GPU; ● represents CPU.

method used the same parameter settings, with 90% as the training set and 10% as the test set. TextCNN, as a non-pre-trained deep learning method, does not require a large number of pre-trained parameters to be computed and achieves the second-best accuracy in less training time (18.6 s on the GPU). Since urban scenes are not as rich as human language, the Bayes method is also a good choice, and the model was trained on the Central Processing Unit (CPU) for only 0.1 s, achieving the third-best result.

We also compared this with the computer vision approach. Similarly, we “stitched together” street images from different viewpoints and used a residual neural network (ResNet) to recognize urban scenes. As there is no vision-language pre-trained model for feature engineering (pre-processing) of the images, costs the most computing time without achieving best results.

Faced with unbalanced data labels (Fig. 8) and high image similarity for some different scenes, the Resnet model suffers from some over-fitting problems. In contrast, scene descriptions obtained with the human attention perspective have higher robustness.

The cross-modal computational process of Step1 necessarily entails a loss of information compared to the direct use of images, but it also brings some benefits in terms of improved generalization and replicability of the model. When directly using visual information, the different countries architectural styles may have an impact on the predictions. Using a highly semantically compressed representation of the image (image caption), the obtained information on urban entities and entity attributes is more conducive to scene recognition.

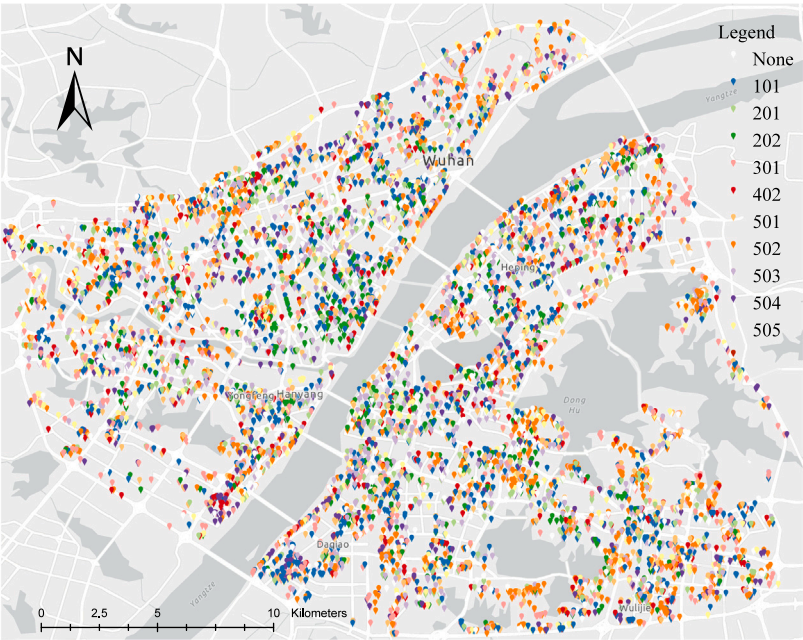


Fig. 9. Prediction results of SFT-BERT model with few-shot learning (80×10 equal number of training samples, 8,123 prediction samples. 7,079 have a clear prediction category. 1,044 sampling locations are multifunctional mixing areas and do not belong to any clear category).

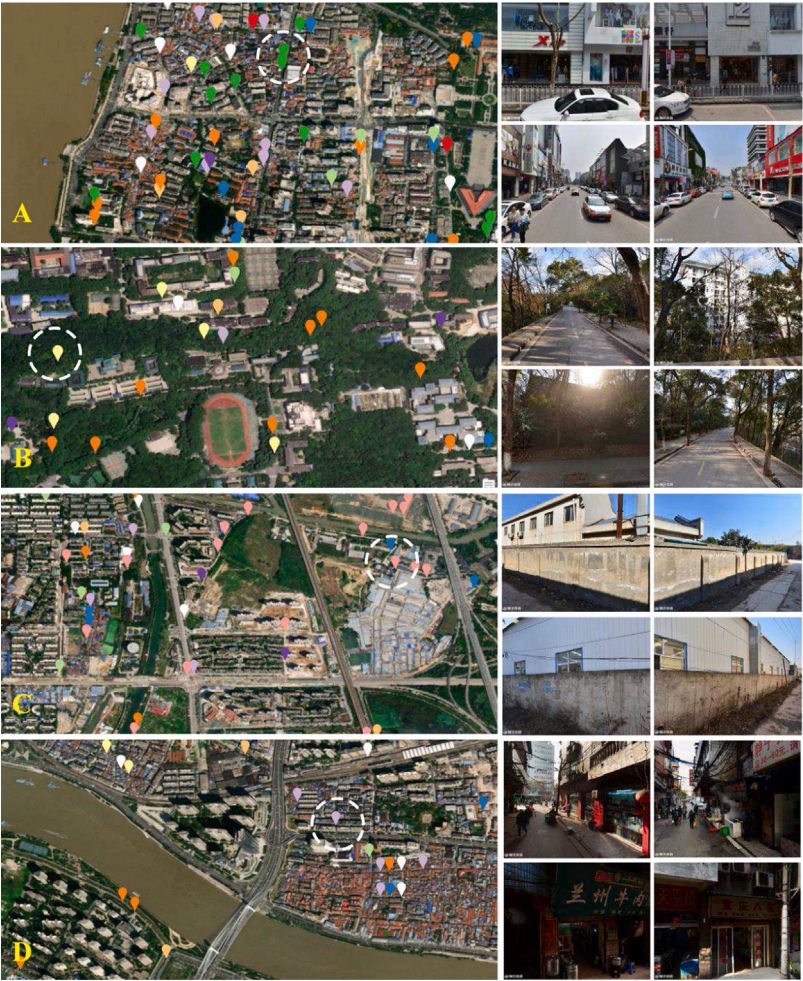


Fig. 10. Comparison of model prediction results with actual street images.

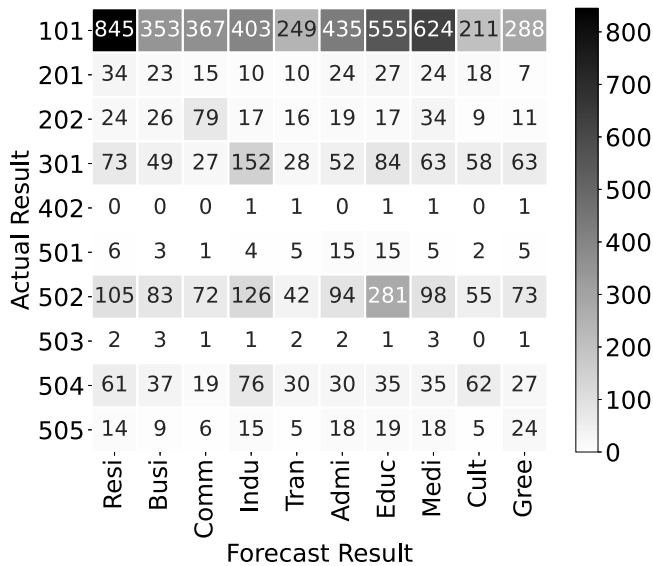


Fig. 11. Confusion matrix of the few-shot learning model.

4.4. Model accuracy testing with few-shot learning and migration experiment

To verify the accuracy of the SFT-Bert model in the few-shot case and to balance the training data distribution of Fig. 8. We randomly select 80 data for each label, a total of 800 data for training, accounting for about 10% of the labeled data. Fig. 9 is the predict result, and unlike Fig. 7 which takes the overall region as the spatial unit, we focus more on the city local scenes. The comparison between our prediction and the actual scene is shown in Fig. 10. The prediction result of Fig. 10 A is consistent with EULUC (Fig. 7), both of which are commercial service category and have obvious characteristics. This type scene is characterized by the abundance of stores on the street sides, full of cars and bustling people. Fig. 10 B predicts the scene as park and green space (504), Fig. 7 predicts the area as educational category (502), from a purely visual point of view, it is difficult to distinguish which category is true. This requires the addition of semantic information such as POI to assist in the judgment. Fig. 10 C is consistent with EULUC, which is in the industrial category (301). The most obvious feature is the large number of industrial plants. Fig. 10 D is predicted as medical category(503), which seems to be a residential area, and the model is less effective in recognizing such scenes without obvious visual features.

We plot the confusion matrix of the few-shot learning results (Fig. 11). Since we use the same number of each category for training, a large number of areas belong to the residential category are classified as others. The EULUC product represents the function of the overall area (Fig. 8), while our prediction is scene level, with higher resolution (Fig. 9). However, our model is not good at distinguishing scenes without visual differences. And it could reduce the number by merging categories with similar visual characteristics (Hu et al., 2021; Xu et al., 2022).

At last, we chose Singapore and New York for migration experiments to verify the geographic replicability of the SFT-BERT model. We sampled Google Street View data of the cities (804 images for 201 locations in New York and 1,608 images for 402 locations in Singapore), and the prediction results are shown in Fig. 12. There are more locations with predicted labels 101,301,502,504, which is similar to the label distribution in Fig. 8. Although our model is trained in Tencent Street View in Wuhan, experiments in Google Street View images in other cities also show high consistency. Note that if we have the label of New City (SG,NY), performing a short time migration

learning will optimize the results even if the classification criteria are not consistent with EULUC.

The advantage of our purely visual solution is that it could ignore the different architectural styles, which is still applicable even migrating to other cities. The disadvantage is that the visual scheme does not contain semantic information, and it is difficult to distinguish categories without obvious visual features.

This section demonstrates the potential of large-scale pre-trained models for GIS-related analysis, using a downstream task of urban scene classification. The advantage is that the user does not need to pay attention to the pre-training process of the model, which has a wide range of application scenarios. The training time shown in Table 3 is clearly acceptable compared to the weeks-long pre-training process for dozens of GPUs. GeoAI tasks often face insufficient data and poor start-up results, which lead to a large amount of upfront human annotation when introducing deep learning algorithms. Large-scale pre-trained models could significantly improve the effectiveness of different GIS task applications and make the models more generalizable. Since STFBERT has been learned from a large corpus, over-fitting could be easily avoided based on few-shot data.

5. Conclusion and discussion

In this paper, a vision language model is introduced to process street view images, which could automatically generate textual interpretations of urban scenes. A loss function for urban scene sensing is also constructed using a semantic representation model.

Our research finds that the pre-trained + fine-tuning approach has an inherent advantage over the TextCNN and MLP models, which train the neural network parameters from scratch. Whereas most of our model's parameters are fixed, the model has already learned a certain amount of feature characterization ability, with an innate knowledge bonus—the larger the trained dataset, the more significant the effect.

Since the BERT pre-trained model was proposed and set at the top of the list of major NLP tasks in 2018, deep learning-based NLP algorithms have gradually transitioned to a paradigm of using large pre-trained models to fine-tune different tasks downstream. This approach leverages large models and a large open-source pre-trained corpus to bring the model's ability to understand natural language up a notch. Cities are major vectors of human activity, generating large amounts of natural language information all the time. Existing GIS research focuses more on private data and lacks universal baseline algorithms and datasets, which is difficult to generalize due to its low information capacity. In addition, we need to consider the spatial heterogeneity. The urban scene prediction also needs to consider the road network topology and regional adjacencies (Zhu et al., 2020), which will be a good direction for optimization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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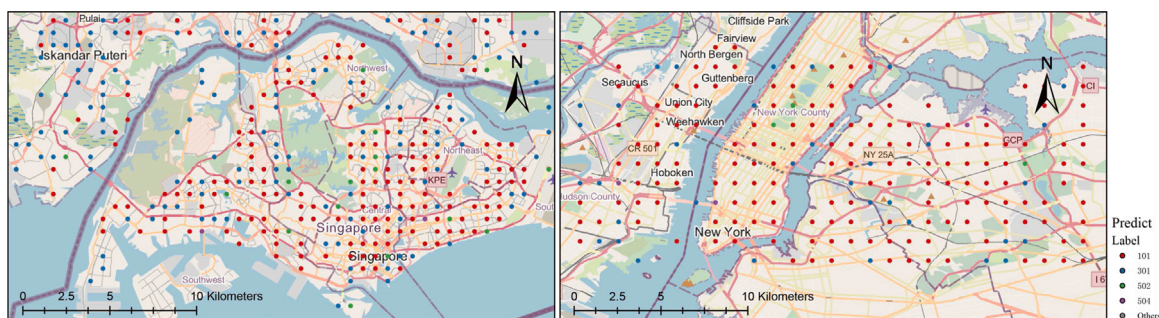


Fig. 12. Prediction Results of SFT-BERT Model in Singapore and New York Area Sampling.

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